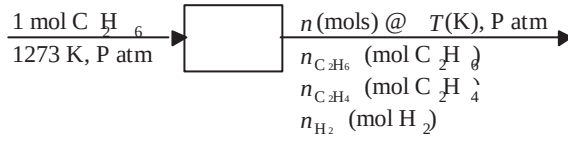


9.35

Basis: 1 mol C₂H₆ fed to reactor

a. $\text{C}_2\text{H}_6 \rightleftharpoons \text{C}_2\text{H}_4 + \text{H}_2$, $K_p = \frac{x_{\text{C}_2\text{H}_4} x_{\text{H}_2}}{x_{\text{C}_2\text{H}_6}} P = 7.28 \times 10^6 \exp[-17,000 / T(\text{K})]$ (1)

Fractional conversion = f (mols C₂H₆ react/mol fed)

$$\left. \begin{array}{l}
 \xi(\text{mol}) = f \\
 n_{\text{C}_2\text{H}_6} = (1-f)(\text{mol C}_2\text{H}_6) \\
 n_{\text{C}_2\text{H}_4} = f(\text{mol C}_2\text{H}_4) \\
 n_{\text{H}_2} = f(\text{mol H}_2) \\
 n = 1 + f(\text{mols})
 \end{array} \right\} \Rightarrow \begin{array}{l}
 x_{\text{C}_2\text{H}_6} = \frac{1-f}{1+f} \frac{\text{mol C}_2\text{H}_6}{\text{mol}} \\
 x_{\text{C}_2\text{H}_4} = \frac{f}{1+f} \frac{\text{mol C}_2\text{H}_4}{\text{mol}} \\
 x_{\text{H}_2} = \frac{f}{1+f} \frac{\text{mol H}_2}{\text{mol}}
 \end{array}$$

$$K_p = \frac{x_{\text{C}_2\text{H}_4} x_{\text{H}_2}}{x_{\text{C}_2\text{H}_6}} P \Rightarrow K_p = \frac{\frac{f^2}{(1+f)^2} P}{\frac{(1-f)}{(1+f)}} = \frac{f^2 P}{(1-f)(1+f)} = \frac{f^2}{1-f^2} P$$

$$(1-f^2)K_p = f^2 P \Rightarrow f = \left(\frac{K_p}{P + K_p} \right)^{1/2} \quad (2)$$

b. References: C₂H₆(g), C₂H₄(g), H₂(g) at 1273 K

Energy balance:

$$\Delta H = 0 \Rightarrow \xi \Delta \hat{H}_r(1273 \text{ K}) + \sum_{\text{out}} n_i \hat{H}_i - \sum_{\text{in}} n_i \hat{H}_i$$

$$(\hat{H}_i)_{\text{in}} = 0 \text{ (inlet temperature = reference temperature)}$$

$$(\hat{H}_i)_{\text{out}} = \int_{1273}^T C_{pi} dT$$

 $\Downarrow$ energy balance

$$f \Delta \hat{H}_r(1273 \text{ K}) + (1-f) \int_{1273}^T (C_p)_{\text{C}_2\text{H}_6} dT + f \int_{1273}^T (C_p)_{\text{C}_2\text{H}_4} dT + f \int_{1273}^T (C_p)_{\text{H}_2} dT = 0$$

 $\Downarrow$ rearrange, reverse limits and change signs of integrals

$$\frac{1-f}{f} = \frac{\Delta \hat{H}_r(1273 \text{ K}) - \int_T^{1273} (C_p)_{\text{C}_2\text{H}_4} dT - \int_T^{1273} (C_p)_{\text{H}_2} dT}{\int_T^{1273} (C_p)_{\text{C}_2\text{H}_6} dT} \quad (3)$$

$$\frac{1-f}{f} = \phi(T) \Rightarrow 1-f = f\phi(T) \Rightarrow f = \frac{1}{1+\phi(T)} \quad (4)$$

9.35 (cont'd)

$$\phi(T) = \frac{145600 - \int_T^{1273} (9.419 + 0.1147T) dT - \int_T^{1273} (26.90 + 4.167 \times 10^{-3} T) dT}{\int_T^{1273} (11.35 + 0.1392T) dT}$$

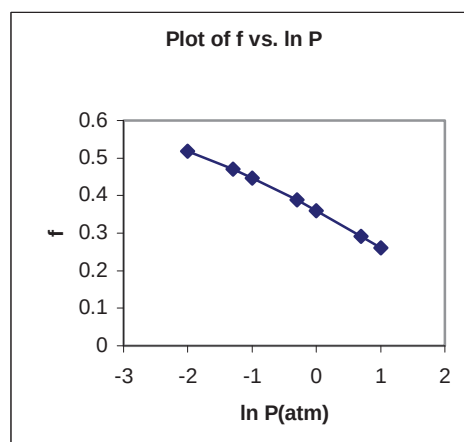
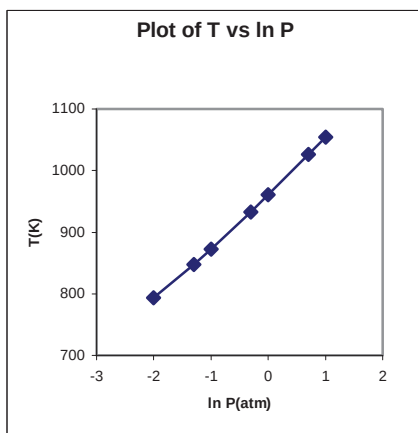
$$\Rightarrow \phi(T) = \frac{3052 + 36.2T + 0.05943T^2}{127240 - 11.3T - 0.0696T^2}$$

c. $\left(\frac{K_p}{1 + K_p} \right)^{1/2} = \frac{1}{1 + \phi(T)} \Rightarrow \left(\frac{K_p}{1 + K_p} \right)^{1/2} - \frac{1}{1 + \phi(T)} = \psi(T) = 0$

$\phi(T)$ given by expression of Part b. $K_p(T)$ given by Eq. (1)

d.

P (atm)	T (K)	f	K _p (atm)	Phi	Psi
0.01	794	0.518	0.0037	0.93152	-0.0001115
0.05	847.4	0.47	0.0141	1.12964	-0.0002618
0.1	872.3	0.446	0.025	1.24028	0.00097743
0.5	932.8	0.388	0.0886	1.57826	3.41E-05
1	960.3	0.36	0.1492	1.77566	4.69E-05
5	1026	0.292	0.4646	2.42913	-2.57E-05
10	1055	0.261	0.7283	2.83692	-7.54E-05



e. C **PROGRAM FOR PROBLEM 9-35
 WRITE (5, 1)
 1 FORMAT ('1', 20X, 'SOLUTION TO PROBLEM 9-35//')
 T = 1200.0
 TLAST = 0.0
 PSIL = 0.0

9.35 (cont'd)

```

C  **DECREMENT BY 50 DEG. AND LOOK FOR A SIGN IN PSI
    DO 10I =1, 20
        CALL PSICAL (T, PHI, PSI)
        IF ((PSIL*PSI).LT.0.0) GO TO 40
        TLAST = T
        PSIL = PSI
        T = T - 50.
10    CONTINUE
40    IF (T.GE.0.0) GO TO 45
        WRITE (3, 2)
        2    FORMAT (1X, 'T LESS THAN ZERO -- ERROR')
            STOP
C  **APPLY REGULA-FALSI
45    DO 50 I = 1, 20
        IF (I.NE.1) T2L = T2
        T2 = (T*PSIL-TLAST*PSI)/(PSIL-PSI)
        IF (ABS(T2-T2L).LT.0.01) GO TO 99
        CALL PSICAL (T2, PHIT, PSIT)
        IF (PSIT.EQ.0) GO TO 99
        IF ((PBIT*PBIL).GT.0.0) PSIL = PSIT
        IF ((PSIT*PSIL).GT.0.0) TLAST = T2
        IF ((PSIT*PSI).GT.0.0) PSI = PSIT
        IF ((PSIT*PSI).GT.0.0) T = T2
50    CONTINUE
        IF (I.EQ.20) WRITE (3, 3)
        3    FORMAT ('0', 'REGULA-FALSI DID NOT CONVERGE IN 20 ITERATIONS')
93    STOP
        END
        SUBROUTINE PSICAL (T, PHI, PSI)
            REAL KF
            PHI = (3052 + 36.2*T + 36.2*T + 0.05943*T**2)/(127240. - 11.35*T
            * - 0.0636*T**2)
            KP = 7.28E6*EXP(-17000./T)
            FBI = SQRT((KP/(1. + KP)) - 1./12. + PHI)
            WRITE (3, 1) T, PSI
            1    FORMAT (6X, 'T =', F6.2, 4X, 'PSI =', E11.4)
            RETURN
        END

```

OUTPUT: SOLUTION TO PROBLEM 9-35

```

T = 1200.00  PSI = 0.8226E + 00
T = 1150.00  PSI = 0.7048E + 00
T = 1100.00  PSI = 0.5551E + 00
T = 1050.00  PSI = 0.3696E + 00
T = 1000.00  PSI = 0.1619E + 00
T = 950.00   PSI = -0.3950E - 01
T = 959.80   PSI = -0.1824E - 02
T = 960.25   PSI = -0.7671E - 04
T = 960.27   PSI = -0.3278E - 05

```

Solution: $T = 960.3 \text{ K}$, $f = 0.360 \text{ mol C}_2\text{H}_6 \text{ reacted/mol fed}$